(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: ES-102 ***Subject: Programming in 'C'***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Answer all the following questions briefly:

a) What is Token? What are the different types of token available in C. **(3)**

b) Enlist the significance of library functions. **(3)**

c) What are actual parameters and formal parameters? Illustrate with example. **(3)**

d) Describe header file and its usage in 'c' programming. **(3)**

e) Differentiate between put(), putchar() and puts(). **(3)**

UNIT-I

Q2 a) Write an algorithm find angle between hour and minutes hand of a clock at a given time. **(7)**

b) Explain the bit wise AND, OR and NOT operation with suitable example. Mention the role of shift operators. **(8)**

Q3 a) What is type conversion? Explain two types of conversion with examples. **(7)**

b) Define pseudocode. Write a pseudocode to find the area of triangle when we know the lengths of all three of its sides. **(8)**

UNIT-II

Q4 a) Write a C program that takes three coefficients (a, b and c) of a quadratic equation: (ax^2+bx+c) as input and compute all possible roots and print them with appropriate messages. **(8)**

b) What is array? Explain the declaration and initialization of one dimensional and two dimensional arrays with an example. **(7)**

Q5 a) Shivam is working on a project which requires returning multiple values from a function. But a return statement can only return single value from a function. How the function should be implemented so that multiple values can be returned by Shivam. Explain with suitable example. **(8)**

b) Write a C program to eliminate multiple spaces from a sentence and make it single. **(7)**

P.T.O.